

1 Criterion validity of heel-mounted inertial 2 sensors for stride-length estimation during 3 single- and dual-task walking in older adults 4 with and without cognitive impairment

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14 ABSTRACT

15 **Background.** Gait metrics such as stride length are sensitive markers of fall and cognitive-decline risk in
16 older adults, yet the reference systems that measure them accurately are limited to short walks in the
17 laboratory. Wearable inertial sensors could extend monitoring into clinical and everyday settings, but their
18 accuracy remains uncertain in the slow, variable gait of older adults.

19 **Methods.** We validated a heel-mounted inertial measurement unit (IMU) system against a GaitRite
20 electronic walkway in 32 community-dwelling older adults (19 in the MoCA-defined cognitively normal
21 group and 13 in the MoCA-defined cognitive impairment group). Participants completed single-task
22 walking and three graded cognitive dual-task walking conditions (serial subtraction by 2s, 3s, and 7s)
23 over a 6.1 m walkway. The pipeline combined bilateral stance detection with retroactive gravity-residual
24 correction, used adaptive-threshold heel-strike and toe-off detection to support bilateral stance validation,
25 and required no per-participant calibration.

26 **Results.** Across 10,472 matched strides retained after walk-level alignment quality control, stride length
27 agreed with the walkway to within 3.9 cm root-mean-square error (RMSE), with a bias of +0.3 cm and an
28 intraclass correlation (ICC) of 0.976. Mean absolute error fell below 4 cm in 31 of 32 participants. Per-
29 condition RMSE ranged from 3.7 to 4.0 cm despite a 16% drop in gait speed under the most demanding
30 condition. Gait speed, stride time, and cadence also agreed closely with the walkway (ICC 0.90–0.96).
31 Participant-level error was modestly higher in the MoCA-defined cognitive impairment group than in the
32 cognitively normal group. Replacing the internal mid-stance integration anchor with GaitRite heel-strike
33 timing did not reduce stride-length RMSE.

34 **Conclusions.** A calibration-free heel-mounted IMU pipeline maintained low stride-length error across
35 graded dual-task walking in older adults. Error was low in both MoCA-defined groups. These findings
36 establish criterion validity for successfully synchronised, supervised straight walkway passes and support
37 further clinical evaluation. Everyday-living validity, other clinical cohorts, and test-retest reliability remain
38 untested.

39 1 INTRODUCTION

40 **Gait metrics such as stride length are sensitive markers of mobility decline, fall risk, and cognitive**
41 **deterioration in older adults.** Greater stride-to-stride variability is a strong predictor of incident falls
42 in older adults (Maki, 1997; Verghese et al., 2009). Slower gait also predicts falls and tracks cognitive
43 decline (Verghese et al., 2009; Montero-Odasso et al., 2012). A meta-analysis reported that a baseline
44 stride-length threshold of 0.64 m or less predicted adverse clinical events and mortality in older adults
45 (Bytyçi and Henein, 2021). Instrumented walkways and motion capture systems measure stride length
46 accurately but restrict assessment to short walks in laboratory settings. This leaves daily-living and clinical
47 contexts without these tools. Wearable inertial measurement units (IMUs) can estimate stride length
48 outside the laboratory, but the quality of these estimates depends on both the measurement setup and the
49 processing method.

50 **Sensor quality, secure attachment, placement, and algorithm design all affect stride-length accuracy**
51 **with foot-mounted IMUs.** The most common approach estimates stride length by integrating acceleration
52 twice. Noise and small sensor errors accumulate during this process and cause the estimated position
53 to drift. A sensor that moves or rotates relative to the shoe can introduce further error because it no
54 longer measures the foot from a consistent position and orientation. The hardware must therefore provide
55 low-noise signals, and the sensor must remain firmly attached (Wahlström and Skog, 2021; Küderle et al.,
56 2022). Zero-velocity updates (ZUPTs) limit drift by resetting the integration during mid-stance, when the
57 foot is flat and motionless (Foxlin, 2005; Wahlström and Skog, 2021). Detecting this period correctly is
58 important because it marks where each stride begins and ends. Adaptive methods that account for each
59 person's walking pattern remain more reliable across walking speeds than a single fixed threshold (Li
60 et al., 2023).

61 **Fixed amplitude thresholds can miss genuine stance phases and misclassify other stationary periods.**
62 During mid-stance the foot is still. Both the sensor's rotational speed (the gyroscope norm, the combined
63 magnitude of triaxial angular velocity) and the moment-to-moment variation in its acceleration then drop
64 close to zero. A threshold detector labels a sample as stance whenever both signals fall below fixed cutoffs
65 (Skog et al., 2010). This works for steady walking, but during slow or irregular walking, such as the
66 shuffling gait of Parkinson's disease, a stance phase may never become quiet enough to cross the cutoff
67 (Foxlin, 2005). The detector then misses the stance and merges two strides into one. During pauses such
68 as standing or turning, the foot is also still, so the detector fires a false stance that shortens the surrounding
69 strides.

70 **Bilateral ZUPT checks each candidate stance against movement of the opposite foot.** During normal
71 walking, one foot lifts from the ground shortly after the opposite foot enters stance. The detector uses this
72 relationship to confirm that a low-motion period represents walking rather than quiet standing, turning,
73 or stopping. In technical terms, it accepts an ipsilateral stance candidate when a toe-off occurs on the
74 contralateral foot within the expected time window (Arens et al., 2021). A recovery step also searches
75 for quiet periods near unmatched opposite-foot toe-offs, allowing the detector to recover stances that a
76 threshold alone missed. This check is particularly relevant to the slow and variable walking of older adults
77 and people with cognitive impairment.

78 **Two gaps remain in the evidence for this approach. The first is the lack of validation under cognitive**
79 **load in older adults.** Foot and shank-mounted IMU validation studies typically examine single-task
80 walking at a steady pace in healthy adults (Sijobert et al., 2015). Cognitive dual-task walking slows gait
81 and increases stride-to-stride variability (Beauchet et al., 2005; Yogev-Seligmann et al., 2008; Hausdorff
82 et al., 2008). Slower, more variable strides widen the range of speeds a pipeline must resolve and make
83 mid-stance phases harder to detect. Older adults, especially those with cognitive impairment, walk
84 more slowly and with greater variability. An IMU pipeline must therefore be validated under these
85 conditions, not solely on steady single-task walking. One heel-mounted IMU study recently assessed
86 intra-session reliability in 101 middle-aged adults and found intra-session reliability but did not validate
87 against a criterion reference (Boutaayamou et al., 2025). Prior validation in pathological cohorts against
88 instrumented treadmills used fixed speeds rather than variable gait speeds (Laidig et al., 2021). Dual-task
89 gait assessment carries clinical relevance because it supports fall-risk screening and cognitive monitoring
90 (Montero-Odasso et al., 2012; Commandeur et al., 2018; Springer et al., 2006). To our knowledge, no
91 prior study has validated gait-event detection and stride-length estimation against a criterion walkway in

92 older adults under graded cognitive load across MoCA-defined cognitive status.

93 **The second gap concerns heel-mounted accuracy in older adults.** The only large heel-specific study
94 examined single-stride variability at multiple shoe-collar positions, including the heel, in healthy adults
95 (mean age 25 years) (Küderle et al., 2022). A prior study demonstrated bilateral validation in post-stroke
96 gait ($n = 8$, wearable-mounted sensors) (Arens et al., 2021). A heel-mounted bilateral ZUPT stance-
97 detection and stride-length pipeline has not been validated against a criterion walkway in older adults
98 walking under graded cognitive load.

99 **Our central contribution is validating two modifications to a calibration-free heel-mounted ZUPT**
100 **pipeline under graded cognitive load.** We validated the pipeline against a GaitRite electronic walkway
101 in 32 community-dwelling older adults. The cohort included 19 participants in the MoCA-defined
102 cognitively normal group and 13 in the MoCA-defined cognitive impairment group. Participants walked
103 under single-task and three graded cognitive dual-task conditions. The pipeline adds contralateral stance
104 validation and retroactive gravity-residual correction to standard per-stride ZUPT integration. It requires
105 no per-participant calibration. This design tests whether low stride-length error persists beyond steady
106 single-task walking.

107 2 MATERIALS & METHODS

108 2.1 Participants

109 **Thirty-two community-dwelling older adults completed the protocol after exclusions.** We recruited
110 33 community-dwelling older adults aged 60 years or older, of whom 32 contributed to the analysis. The
111 analysed cohort comprised 8 male and 24 female participants, with mean age 70.8 ± 7.8 years (range
112 60–83), mean height 167.4 ± 7.6 cm, and mean body mass 73.0 ± 13.2 kg (per-participant values in
113 Supplementary Table S1). Supplementary Table S2 reports the two participant-condition records whose
114 paired source data were unavailable. We formed two screening groups using the Montreal Cognitive
115 Assessment (MoCA) (Nasreddine et al., 2005). The groups included 19 cognitively normal participants
116 (CN, MoCA ≥ 26) and 13 participants meeting the MoCA criterion for cognitive impairment (CI, MoCA
117 14–25). These labels denote MoCA-defined screening groups rather than clinical diagnoses. Inclusion
118 criteria were a MoCA score of 14 or higher and the ability to walk 20 m independently without a mobility
119 aid. The University of Victoria Human Research Ethics Board approved the protocol (H22-00451), and
120 all participants provided written informed consent.

121 2.2 Instrumentation

122 **Two heel-mounted IMUs and a pressure-sensitive walkway recorded each walk concurrently.** We
123 attached two wireless IMUs (Opal, APDM Inc., Portland, OR) to the posterolateral heel of each shoe,
124 adjacent to the calcaneus, using a 3D-printed housing fastened with a clip. Prior validation studies show
125 the Opal sensor measures spatiotemporal gait parameters accurately against treadmill and instrumented-
126 walkway references in healthy adults and older adults (Washabaugh et al., 2017; Morris et al., 2019).
127 Each IMU recorded triaxial accelerometer (± 16 g) and gyroscope ($\pm 2000^\circ/\text{s}$) data at 256 Hz.

128 A pressure-sensitive electronic walkway (GaitRite, model Platinum, CIR Systems Inc., Franklin, NJ)
129 served as the criterion reference. The walkway has an active sensing area of $594 \text{ cm} \times 61 \text{ cm}$ with 1.27 cm
130 square sensors that record heel-on to toe-off stance events for each foot at 120 Hz. GaitRite gait parameters
131 show excellent concurrent validity against three-dimensional motion capture, with ICC 0.92–0.99 across
132 walking speed, cadence, step length, and step time (Webster et al., 2005). We synchronised the walkway
133 and IMU clocks using a GaitRite Pulse (GR Pulse) signal generated by a custom-written LabVIEW
134 program. The GR Pulse marked the start and end of each walkway pass on the LabVIEW clock. Within
135 each participant-condition pair, we fitted a linear clock correction between GR Pulse rising edges and
136 corresponding IMU activity onsets across up to seven walks. We then estimated a residual constant offset
137 within each walk by aligning adaptive-detector IMU heel strikes with GaitRite heel strikes.

138 2.3 Protocol

139 **Each participant completed four walking conditions, one single-task and three graded dual-task**
140 **conditions.** Single-task walking served as the baseline, followed by three dual-task conditions of graded
141 cognitive difficulty. Participants counted backwards aloud by 2s, 3s, and 7s from a random three-digit
142 starting number while walking. Each condition comprised seven trials of two passes (out and back) over
143 the GaitRite mat at a self-selected comfortable pace. This yielded 14 walks per condition and 56 walks
144 per participant across the four reported conditions. Participants walked beyond both ends of the mat to
145 allow acceleration and deceleration off the recording surface. Participants wore a safety belt, and a spotter
146 accompanied them. We provided rest periods between conditions as needed.

147 2.4 Signal preprocessing

148 **We bandpass-filtered the gyroscope signal and aligned the left-right sign convention before further**
149 **processing.** We first bandpass-filtered raw gyroscope signals (0.5–10 Hz, fourth-order Butterworth) to
150 remove low-frequency drift and high-frequency noise. We then inverted the left heel gyroscope Z-axis
151 to match the right heel sign convention (positive = plantarflexion angular velocity in the sagittal plane).
152 We left the triaxial accelerometer signals unfiltered for the ZUPT integration pipeline. The per-stride
153 zero-velocity and gravity-residual corrections addressed integration drift directly, so we did not add an
154 accelerometer filter whose cutoff would require further tuning.

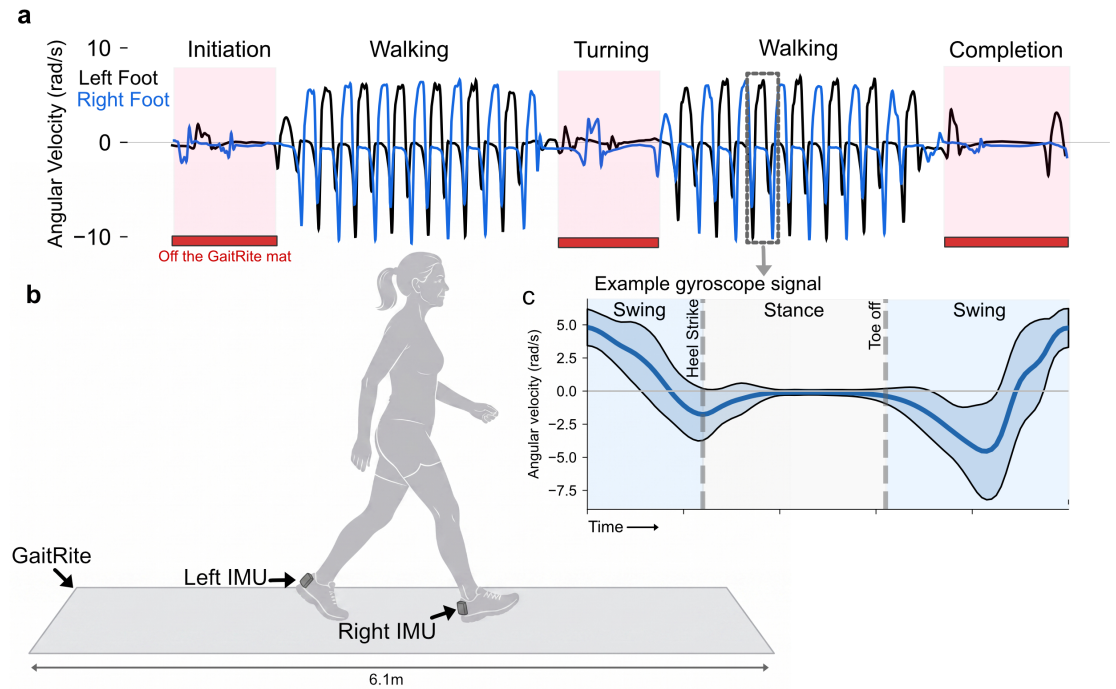


Figure 1. Experimental setup and gyroscope processing. (a) Filtered sagittal-plane signals from the left (black) and right (blue) heel IMUs during one out-and-back trial. Labels mark activity phases, red bars mark off-mat intervals, and the box is expanded in panel c. (b) Heel IMUs clipped to the posterolateral shoe using 3D-printed housings during GaitRite walking. (c) Mean gyroscope signal over a heel-strike-defined gait cycle, pooled across participants and conditions. Shading shows ± 1 SD, dashed lines mark heel strike and toe off, and blue marks swing.

2.5 Gait event detection

We detected heel strikes and toe-offs on each foot using thresholds that adapted to the participant's walking pattern. A fixed cutoff can miss events when stride amplitudes change with walking pace. We therefore adapted an established sagittal-plane gyroscope detector (Salarian et al., 2004) using signal-derived thresholds (Greene et al., 2010). The detector estimated stride frequency from the repeating pattern of the filtered gyroscope signal. Heel-strike and toe-off peak thresholds used 35% and 30% of the negative-peak amplitude interquartile range, respectively. Cadence scaled both thresholds. The peak threshold could not fall below 15% of the maximum absolute signal amplitude or 0.5 rad/s. Positive swing confirmation used 25% of the maximum absolute amplitude with a 1.0 rad/s floor. The detector then classified negative signal peaks according to the pattern that followed within 80–220 ms. A rapid rise into swing identified toe-off, while a near-zero segment identified heel strike (Fig. 1c). Toe-offs supported bilateral stance validation, and both events were compared with GaitRite for the timing analysis. Stride time for paired analyses came from consecutive stance midpoints because these stationary anchors also bounded each integrated displacement. We analysed discrete heel-strike timing separately against GaitRite initial contacts.

2.6 Ipsilateral stance phase detection with contralateral validation

Stance-phase detection runs in two steps. A low-motion detector first finds candidate stance windows on one foot. A toe-off on the opposite foot then confirms each candidate. Here, ipsilateral means the foot being evaluated and contralateral means the opposite foot. The resulting stance mask anchors the ZUPT integration. We define stride boundaries at the midpoints of consecutive stance phases because the foot is stationary there, providing a stable zero-velocity reference.

Step 1. Ipsilateral low-motion detection. The detector evaluated the gyroscope magnitude and accelerom-

177 eter variability from the same heel sensor in a sliding 40 ms window (Skog et al., 2010; Foxlin, 2005). A
178 sample qualified as a stance candidate when both signals fell below adaptive thresholds calculated for
179 each participant and walking condition. The gyroscope criterion identified low rotation during foot-flat
180 contact, while the accelerometer criterion rejected brief low-rotation periods during swing. Algorithm 1
181 lists the thresholds, limits, and recovery window used in the analysis.

182 *Step 2. Opposite-foot confirmation.* The detector accepted a stance candidate when a toe-off on the
183 opposite foot fell within 150 ms of either boundary of the candidate segment (Arens et al., 2021).
184 Candidates without a matching toe-off were rejected. An unmatched toe-off triggered a recovery search
185 for a quiet period on the other foot. The detector also removed very short stance segments, joined segments
186 separated by brief gaps, and checked unusually long strides for a missed stance. These steps reduced false
187 detections during stops and prevented two strides from being merged.

188 *Fallback to threshold-only detection.* The pipeline reverts to ipsilateral threshold-only stance detection
189 whenever contralateral validation returns an implausibly low stance count. We defined this as fewer than
190 three validated segments or fewer than 50% of the threshold-only segments when the threshold-only
191 detector found at least ten. This guard fired in 11 of 256 subject-condition-foot units (4.3%). Eight of the
192 11 units came from a single participant whose contralateral toe-off detection failed in every condition.
193 Most paired strides therefore used contralateral validation, while the fallback prevented loss of otherwise
194 usable data when contralateral event detection failed.

195 2.7 ZUPT stride length integration

196 **We estimated stride length by double-integrating the accelerometer signal between consecutive**
197 **stance midpoints.** During each stance phase the foot rests on the ground at zero velocity. The stance
198 mask from Section 2.6 flags these zero-velocity intervals, which are used to correct the drift that double
199 integration would otherwise produce (Foxlin, 2005; Skog et al., 2010). We estimated gyroscope and
200 accelerometer biases once per walking bout, defined as a run of consecutive stance phases separated by
201 no gap longer than 3 s. The stance phase supplies the reference, because true angular rate is zero and
202 true acceleration equals gravity. We bias-corrected the signals with these estimates before integration,
203 removing the slow sensor offsets that would otherwise accumulate as drift.

204 2.7.1 Orientation tracking

205 **We tracked sensor orientation with a Mahony attitude-and-heading reference system.** This filter
206 combines gyroscope and accelerometer measurements to estimate how the sensor is oriented relative
207 to gravity (Mahony et al., 2008). We used a higher correction gain during stance ($K_p = 1.0$), when the
208 stationary foot provides a reliable gravity reference, and a lower gain during swing ($K_p = 0.05$), when
209 movement affects the accelerometer signal. The integral gain remained at $K_i = 0.02$. We reduced the
210 proportional accelerometer correction gain during swing to limit contamination by dynamic acceleration.
211 This follows the movement-gating rationale of foot-mounted filters that disable accelerometer-based
212 orientation updates during swing (Hannink et al., 2017).

213 2.7.2 Gravity removal, double integration, and stride length extraction

214 **We integrated the gravity-removed acceleration twice between consecutive stance midpoints to**
215 **estimate stride length.** We rotated the bias-corrected accelerometer signal into the navigation frame
216 using the Mahony-estimated orientation, subtracted gravity, and double-integrated to obtain velocity and
217 position (Foxlin, 2005). Because the foot is momentarily stationary at each stance boundary, its true
218 velocity is zero. We applied a linear zero-velocity correction that distributes the accumulated velocity error
219 backward across the preceding swing samples and then resets the velocity to that known zero, cancelling
220 the drift built up over the stride. Each stride integrates independently with position and velocity reset at
221 each stance midpoint, which bounds drift to a single swing phase. Stride length is the magnitude of the
222 horizontal component of the 3D displacement between consecutive stance midpoints. We accepted strides
223 with duration 0.4–2.0 s, length 0.3–2.5 m, and minimum gyroscope amplitude 3.0 rad/s, a low floor that
224 excludes near-stationary segments rather than genuine strides.

225 2.7.3 Retroactive gravity-residual correction

226 **We corrected the acceleration residual that remained after orientation tracking and gravity removal.**
227 The code refers to this step as bidirectional smoothing. Small orientation errors during swing leave
228 a horizontal acceleration residual after gravity removal, which can inflate the integrated stride length
229 (Nilsson et al., 2012; Mahony et al., 2008). The correction reduced this positive offset in the current
230 cohort (Supplementary Table S4).

231 We corrected this residual retroactively after each swing phase. At each stance onset we measured
232 the residual. We skipped the first 10 ms to avoid the heel-strike impact transient, then averaged the
233 navigation-frame acceleration over the next 20 ms. We then subtracted a linear ramp of this residual
234 through the preceding swing samples, scaled by a gain of 0.5:

$$\mathbf{a}_{\text{corrected}}(t) = \mathbf{a}_{\text{nav}}(t) - 0.5 \cdot \frac{t}{T_{\text{swing}}} \cdot \mathbf{r} \quad (1)$$

235 where t is time from swing onset, T_{swing} is swing duration, and $\mathbf{r}$ is the gravity residual at the next stance
236 onset. We selected the gain of 0.5 empirically in this cohort because no prior work specifies a value for
237 this correction. This selection was not independent of the validation sample. The correction operates per
238 stride and requires no external calibration. A sensitivity analysis across gains from 0.3 to 0.7 showed
239 pooled RMSE values from 3.83 to 4.10 cm (Supplementary Table S7).

240 2.7.4 Algorithm summary

241 **Algorithm 1 summarises the complete processing sequence.** Step 1 builds the stance mask using
242 low-motion detection and opposite-foot confirmation. Step 2 tracks orientation, removes gravity, integrates
243 each stride, and applies the retroactive gravity-residual correction before resetting velocity. Contralateral
244 stance validation and the gravity-residual correction distinguish the tested pipeline from a standard ZUPT
245 implementation (Foxlin, 2005; Skog et al., 2010).

246 2.8 GaitRite stride matching

247 **We paired each ZUPT stride with its GaitRite counterpart using clock alignment and stride-level**
248 **checks.** The GR Pulse correction placed each walk on the IMU timeline, and heel-strike alignment
249 supplied a final offset within each walk. We accepted a walk-level alignment only when at least 60% of
250 the expected GaitRite heel strikes had an IMU match within 150 ms. This required most reference events
251 to match while allowing one or two missed detections in a short pass. Walks without a valid alignment
252 were classified as synchronisation failures and did not enter paired stride or event-agreement analyses.
253 We paired strides from the same foot when the IMU and GaitRite stride midpoints were each other's
254 closest match. The allowable time difference adapted to stride duration and never exceeded 0.5 s. We
255 retained GaitRite strides with lengths of 0.30–2.00 m and times of 0.40–1.80 s. The matched IMU length
256 and time also had to fall between 50% and 150% of the corresponding reference value. We call this the
257 stride-pair mismatch filter. It removed likely pairing errors and GaitRite records in which a missed heel
258 strike produced an apparent double stride.

259 We retained only strides that fell completely within a GaitRite walk window. This removed turns, off-
260 mat acceleration and deceleration, mat-entry transitions, and rest periods. Participants began walking
261 approximately 1 m before the mat, so gait-initiation strides were not part of the synchronised comparison.
262 Partial steps at either end of the mat were removed before matching.

263 **The final analysis included 10,472 paired strides from 32 participants.** We excluded one of the 33
264 enrolled participants because the required heel-mounted IMU recordings were unavailable. We retained
265 every reported participant-condition record that had the required source data and at least one stride pair
266 after the common matching rules. We did not exclude records based on match count or stride-length error
267 because both are study outcomes. GR Pulse records were unavailable for ID_186 L2 and ID_195 L2, so
268 these two conditions could not enter the paired analysis (Supplementary Table S2). Of 1,764 available
269 walks, 1,669 passed alignment QC and 95 were classified as synchronisation failures. Failures occurred in
270 14 single-task, 13 serial-2, 19 serial-3, and 49 serial-7 walks. We also excluded three individual walks
271 for documented data-quality problems described in Supplementary Section S2. After stride-level checks,

Algorithm 1: Bilateral ZUPT stride-length estimation.

Input : Ipsilateral 3-axis accelerometer (*accel*) and gyroscope (*gyro*)
Contralateral toe-off times (*contra_to*)
Sampling rate (*fs*)

Output : Stride lengths (*stride_lengths*)

```
/* Step 1. Stance detection with contralateral validation */
1 gyro_norm = magnitude of gyro at each sample
2 accel_var = 40 ms sliding variance of accel
3 gyro_thresh = clip(15th percentile of gyro_norm,
4 0.3, 2.0 rad/s)
5 accel_thresh = clip(25th percentile of accel_var,
6 0.2, 5.0 (m/s2)2)
7 candidates = samples where gyro_norm < gyro_thresh and accel_var <
  accel_thresh
8 for each candidate segment do
9   if any contra_to event falls within  $\pm 150$  ms of the segment boundary then
10    | accept segment as stance
11   else
12    | reject segment
13   end
14 end
15 recover missed stances near unmatched contra_to using relaxed gyroscope-only threshold
  (25th percentile, 0.5 rad/s floor)
16 reject stance segments < 30 ms, merge segments separated by < 75 ms, split strides > 1.8 s at a
  mid-stride quiet period
/* Step 2. Per-stride integration with retroactive
  gravity-residual correction */
17 estimate gyro_bias and accel_bias from stance samples of each walking bout
18 for each stride between consecutive stance midpoints do
19   track orientation with Mahony AHRS (gain 1.0 during stance, 0.05 during swing)
20   rotate (accel - accel_bias) into the navigation frame, subtract gravity
21   measure gravity residual at the next stance onset
22   subtract a linear ramp of that residual through the preceding swing (gain 0.5)
23   integrate corrected acceleration  $\rightarrow$  velocity, distribute the accumulated velocity error
    linearly back across the swing, then set velocity=0 at stance ends
24   integrate velocity  $\rightarrow$  position
25   stride_length = horizontal distance between stride start and stride end positions
26   reject stride unless duration 0.4–2.0 s, length 0.3–2.5 m, peak gyro_norm > 3.0 rad/s
27 end
28 return stride_lengths
```

272 the final pool contained 2,544 single-task, 2,618 serial-2, 2,770 serial-3, and 2,540 serial-7 strides. The
273 conditions included 32, 30, 32, and 32 participants, respectively.

274 2.9 Outcome measures and statistical analysis

275 **The primary outcome was pooled stride-level RMSE between IMU and GaitRite stride length.**
276 Agreement outcomes included signed bias, SD of the error, 95% limits of agreement, MAE, ICC(2,1), and
277 proportional bias (Bland and Altman, 1986; Shrout and Fleiss, 1979). We also reported the percentage
278 of strides within 5 and 10 cm and calculated RMSE and MAE at the participant level. Event-detection
279 outcomes were miss rate, timing bias, and RMSE against GaitRite heel strikes and toe-offs. A missed
280 event had no IMU match within 150 ms. To assess stride-to-stride consistency, we calculated stride-length
281 coefficient of variation for each participant and condition and compared the IMU and GaitRite values
282 using bias, RMSE, and ICC(2,1).

283 We evaluated gait parameters other than stride length using RMSE, bias, and ICC(2,1). We also performed
284 two design comparisons. First, we ran the full stride-length pipeline with bilateral and ipsilateral threshold-
285 only stance detection while keeping all other parameters identical. We compared matched stride pairs,
286 RMSE, bias, and ICC between methods. Second, we compared the internal mid-stance integration
287 boundaries with GaitRite-derived heel-strike boundaries. For this comparison, we used RMSE, bias,
288 MAE, SD of the error, 95% limits of agreement, and ICC(2,1).

289 We tested the effect of cognitive load on stride-length accuracy with a repeated-measures analysis of
290 variance (ANOVA) on participant-level mean absolute error, restricted to the 30 participants with valid
291 data in all four conditions. We applied the Greenhouse–Geisser correction when Mauchly’s test indicated a
292 sphericity violation. We then ran pairwise condition comparisons with Bonferroni correction and focused
293 reporting on each dual-task condition against single-task walking. We report generalised eta-squared (η_G^2)
294 for the overall ANOVA. We compared participant-level MAE between the MoCA-defined groups with a
295 two-sided Mann–Whitney test. This subgroup analysis was exploratory and was not adjusted for multiple
296 comparisons. We described the stride-level relationship between gait speed and absolute error using linear
297 regression and reported R^2 without using the regression for participant-level inference. Participant MAE
298 confidence intervals in Fig. 3 used 2,000 walk-cluster resamples. We ran all analyses in Python 3.13 with
299 pingouin 0.6.1, statsmodels 0.14.6, and scipy 1.17. We selected and reported reliability and agreement
300 measures following the Guidelines for Reporting Reliability and Agreement Studies (GRRAS) (Kottner
301 et al., 2011).

302 **We assessed cohort-level accuracy with a participant-clustered bootstrap.** We resampled the 32
303 participants with replacement and recomputed RMSE, MAE, bias, and ICC on the resulting stride pool.
304 We repeated this process 10,000 times to obtain 95% percentile confidence intervals. The sample size was
305 based on participant availability rather than a prospective power calculation.

306 **We ran two sensitivity analyses.** First, we repeated the accuracy analysis without the stride-pair mismatch
307 filter defined above. All other matching and filtering rules remained unchanged. This analysis tested
308 whether the filter affected the reported accuracy. Second, we varied the retroactive gravity-residual
309 correction gain from 0.3 to 0.7 in increments of 0.1 while keeping the rest of the pipeline unchanged. We
310 compared retained pairs, RMSE, MAE, bias, and ICC(2,1). These same-cohort analyses assess robustness
311 to analysis choices but do not replace independent validation.

3 RESULTS

3.1 Gait event detection accuracy

The adaptive-threshold detector matched GaitRite heel strikes within 33 ms RMSE and toe-offs within 39 ms RMSE. Heel-strike miss rates ranged from 0.0% to 0.4% and toe-off miss rates from 2.8% to 3.8% across conditions (Table 1). Heel-strike RMSE was 27–33 ms and toe-off RMSE was 33–39 ms. A missed event denotes a heel strike or toe-off that the detector could not localise within a stride the pipeline had identified.

Table 1. Adaptive-threshold gait-event accuracy. HS = heel strike; TO = toe off. Miss rate is the percentage of GaitRite events unmatched within 150 ms. Bias = mean IMU-minus-GaitRite timing error.

Condition	Event	Miss rate (%)	<i>N</i> matched	Bias (ms)	RMSE (ms)	MAE (ms)
Single-task	HS	0.0	3749	−0.7	26.8	16.7
	TO	2.8	3698	+3.9	32.8	22.7
Serial 2s	HS	0.2	3650	−0.2	31.2	18.8
	TO	3.2	3586	−0.7	37.8	25.9
Serial 3s	HS	0.4	3858	+0.1	32.4	19.8
	TO	3.6	3760	0.0	38.7	26.6
Serial 7s	HS	0.4	3554	+0.1	32.6	20.1
	TO	3.8	3499	+1.7	39.3	27.3

3.2 Stance detection and event-timing comparisons

Bilateral validation recovered more reference strides with similar accuracy. On the GaitRite-labelled strides, bilateral validation recovered 10,472 matched pairs compared with 9,708 for threshold-only detection (Table 2). The two methods differed by 0.1 cm RMSE and both returned an ICC of 0.976.

Substituting GaitRite heel-strike timing for the internal mid-stance anchor did not reduce stride-length error. We re-ran the ZUPT integrator with GaitRite-derived heel-strike times as displacement endpoints on the 10,459 strides matched in both anchoring schemes. RMSE rose from 3.9 to 4.1 cm and ICC fell from 0.976 to 0.973 (Table 2).

Table 2. Pipeline design comparisons. The blocks compare stance detection and integration anchors on their respective shared stride sets, so *N* differs. All other pipeline steps were held constant. Bias = IMU minus GaitRite.

Variant	<i>N</i> pairs	RMSE (cm)	Bias (cm)	MAE (cm)	ICC(2,1)
<i>Stance detection</i>					
Bilateral (canonical)	10,472	3.9	+0.3	2.5	0.976
Ipsilateral (threshold-only)	9,708	3.8	−0.3	2.4	0.976
<i>Gait-event timing anchor</i>					
Mid-stance (canonical)	10,459	3.9	+0.3	2.5	0.976
GaitRite heel strike	10,459	4.1	+1.1	2.8	0.973

3.3 Stride length accuracy

The pipeline measured stride length with 3.9 cm RMSE and 2.5 cm MAE. The pooled analysis included 10,472 matched stride pairs and returned an ICC of 0.976. Bias was +0.3 cm and the SD of error was 3.9 cm (Table 3, Fig. 2). The scatter plot shows agreement across the full stride-length range (Fig. 2a). The Bland-Altman plot shows a small mean bias and a weak trend across the range (Fig. 2b; proportional-bias slope 0.036). Per-condition RMSE ranged from 3.7 to 4.0 cm, and per-condition ICC ranged from 0.971 to 0.977. The conditions included 32, 30, 32, and 32 participants (Table 3). The error fell within ± 5 cm for 87% of stride pairs and within ± 10 cm for 97%. Proportional bias was weak (slope

0.036, $R^2 = 0.026$). Relative to the mean GaitRite stride length of 1.20 m, pooled RMSE was 3.2% and MAE was 2.1%.

Bias was positive across all four conditions and error distributions overlapped (Table 3). Participant-level MAE ranged from 1.6 to 4.7 cm, with 31 of 32 participants below 4 cm (Fig. 3). Mean per-participant RMSE was 3.8 cm and ranged from 2.1 to 7.2 cm.

When we disabled the retroactive gravity-residual correction on the same matched strides, pooled RMSE and bias increased (Supplementary Table S4).

Table 3. Bilateral ZUPT stride-length accuracy by condition. n = participants; N = matched stride pairs; bias = mean IMU-minus-GaitRite error; LoA = Bland–Altman 95% limits of agreement.

Condition	n	N pairs	Bias (m)	RMSE (m)	MAE (m)	95% LoA (m)	ICC(2,1)
Single-task	32	2,544	+0.007	0.040	0.027	[−0.069, +0.084]	0.973
Serial 2s	30	2,618	+0.001	0.037	0.024	[−0.072, +0.073]	0.977
Serial 3s	32	2,770	+0.000	0.039	0.024	[−0.077, +0.077]	0.975
Serial 7s	32	2,540	+0.003	0.039	0.025	[−0.073, +0.078]	0.971
Pooled	32	10,472	+0.003	0.039	0.025	[−0.073, +0.078]	0.976

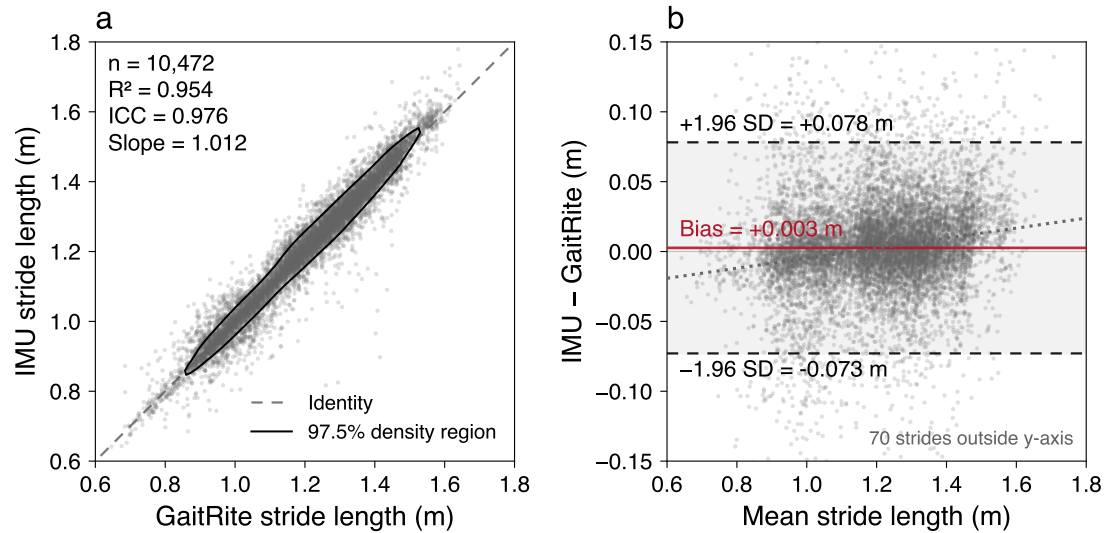


Figure 2. Stride length agreement between IMU and GaitRite. (a) All 10,472 matched strides. The dashed line marks identity, and the contour encloses the densest 97.5% of points. Linear fit $R^2 = 0.954$, slope = 1.012, and ICC(2,1) = 0.976. (b) Bland–Altman plot. The red line marks mean bias (+0.003 m), dashed lines mark the 95% limits of agreement (−0.073 to +0.078 m), and the dotted line shows proportional bias (slope = 0.036).

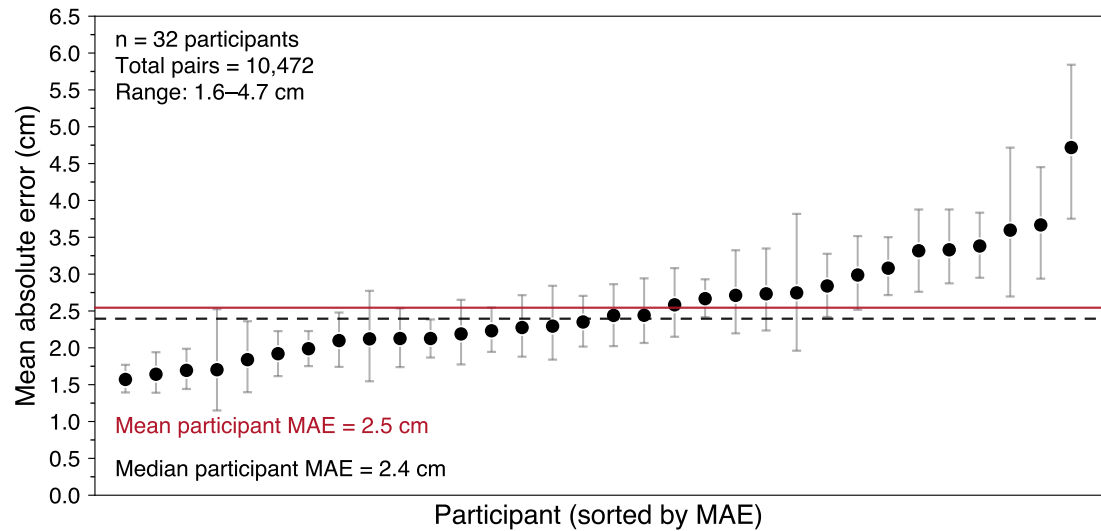


Figure 3. Participant stride-length accuracy across conditions ($n = 32$). Dots show participants sorted by MAE with 95% CIs. The solid red and dashed grey lines mark the mean (2.5 cm) and median (2.4 cm), respectively.

3.3.1 Effect of cognitive load and cognitive status on stride length accuracy

Accuracy did not differ across walking conditions, but participant-level error was modestly higher in the MoCA-defined CI group. The repeated-measures ANOVA on participant-level MAE included the 30 participants with data in all four conditions. It found no evidence of a condition effect ($F(2.35, 68.27) = 0.67$, Greenhouse–Geisser corrected $p = .541$, $\eta_G^2 = 0.016$). No comparison between single-task and a dual-task condition was significant after Bonferroni correction. Absolute error was also weakly related to gait speed ($R^2 = 0.002$). Mean signed bias was below 1 cm in every condition (Table 3).

Stratified by MoCA-defined cognitive status, RMSE was 3.65 cm in CN participants ($n = 19$, 6,139 strides) and 4.14 cm in CI participants ($n = 13$, 4,333 strides). Median per-participant MAE was 2.28 cm in the CN group and 2.84 cm in the CI group (Mann–Whitney $U = 71$, $p = .046$). Error remained low in both screening groups, although participant-level error was modestly higher in the CI group.

3.4 Robustness of accuracy estimates

Participant-clustered bootstrap confidence intervals show stable cohort-level accuracy under participant resampling. We resampled the 32 participants with replacement 10,000 times and recomputed accuracy metrics on each resampled cohort's stride pool. RMSE was 3.86 cm (95% CI [3.51, 4.23]) and MAE was 2.50 cm (95% CI [2.30, 2.70], Supplementary Table S5). Bias remained small (+0.26 cm, 95% CI [−0.08, +0.63]). ICC was 0.976 (95% CI [0.964, 0.981]).

Typical absolute error remained similar without the mismatch filter. Removing the stride-pair mismatch filter added 22 strides and increased RMSE from 3.86 to 5.49 cm, while MAE changed from 2.50 to 2.65 cm (Supplementary Table S6). The larger RMSE change reflects a small number of apparent missed or merged stride events. Across correction gains from 0.3 to 0.7, RMSE ranged from 3.83 to 4.10 cm and was 3.86 cm at the selected gain of 0.5 (Supplementary Table S7).

3.5 Other spatiotemporal gait parameters

The IMU reproduced the graded gait changes caused by cognitive load. GaitRite gait speed dropped 16.2% from single-task to serial 7s (1.19 to 0.99 m/s), stride time lengthened (1.09 to 1.22 s), and cadence fell (111 to 100 steps/min, Table 4). Mean GaitRite stride length also differed across conditions in the 30 participants with complete data ($F(1.39, 40.17) = 88.99$, Greenhouse–Geisser corrected $p < .001$). IMU

values tracked the reference values across all four conditions. Pooled ICC(2,1) was 0.961 for gait speed, 0.905 for stride time, and 0.902 for cadence.

Stride-length variability showed only moderate pooled agreement. Per-participant stride-length CV was 3.58% for the IMU and 3.08% for GaitRite. The pooled bias was +0.50 percentage points and ICC(2,1) was 0.576 (Table 4).

Table 4. Spatiotemporal agreement by condition across 10,472 matched strides. GaitRite and IMU values are mean (SD), and bias = IMU minus GaitRite. Stride-length CV is calculated within participant and reported as the cohort mean.

Parameter	Condition	GaitRite	IMU	Bias	RMSE	ICC(2,1)
Gait speed (m/s)	Single-task	1.186 (0.207)	1.189 (0.215)	+0.002	0.070	0.945
	Serial 2s	1.039 (0.231)	1.043 (0.237)	+0.004	0.064	0.963
	Serial 3s	0.992 (0.229)	0.994 (0.233)	+0.002	0.062	0.964
	Serial 7s	0.994 (0.196)	0.998 (0.202)	+0.004	0.063	0.949
	Pooled	1.051 (0.230)	1.054 (0.236)	+0.003	0.065	0.961
Stride time (s)	Single-task	1.088 (0.097)	1.093 (0.110)	+0.005	0.063	0.816
	Serial 2s	1.181 (0.160)	1.177 (0.158)	−0.004	0.071	0.900
	Serial 3s	1.209 (0.168)	1.207 (0.168)	−0.002	0.071	0.911
	Serial 7s	1.218 (0.165)	1.216 (0.165)	−0.002	0.073	0.903
	Pooled	1.175 (0.159)	1.174 (0.160)	−0.000	0.070	0.905
Cadence (steps/min)	Single-task	111.2 (9.9)	110.9 (11.1)	−0.33	6.18	0.827
	Serial 2s	103.3 (12.8)	103.7 (13.2)	+0.36	5.85	0.899
	Serial 3s	101.1 (13.1)	101.3 (13.5)	+0.21	5.57	0.912
	Serial 7s	100.2 (12.5)	100.4 (12.9)	+0.18	5.67	0.900
	Pooled	103.9 (12.9)	104.0 (13.4)	+0.11	5.82	0.902
Stride-length CV (%)	Single-task	2.69	3.24	+0.56	1.12	0.504
	Serial 2s	3.18	3.39	+0.21	0.82	0.827
	Serial 3s	3.34	3.89	+0.55	1.74	0.518
	Serial 7s	3.12	3.79	+0.66	1.60	0.489
	Pooled	3.08	3.58	+0.50	1.38	0.576

374 4 DISCUSSION

375 **This validation of bilateral heel-mounted IMUs across two MoCA-defined cognitive-status groups**
376 **yielded four main findings.** First, bilateral heel-mounted IMUs matched GaitRite stride length to within
377 3.9 cm pooled RMSE with ICC 0.976 across 10,472 paired strides in older adults walking under single-
378 and dual-task conditions. Second, bilateral validation recovered more GaitRite reference strides than
379 threshold-only detection with similar accuracy. Third, per-condition RMSE ranged from 3.7 to 4.0 cm
380 despite a 16% drop in gait speed. Fourth, error remained low in both MoCA-defined groups, although
381 participant-level error was modestly higher in the CI group.

382 **Our main advance is criterion validation of a practical heel-mounted system across graded cognitive**
383 **load in older adults.** Heel-mounted sensors remain uncommon in gait-validation studies (Lin et al.,
384 2025; Küderle et al., 2022). Previous studies reported low-centimetre errors at other foot and lower-leg
385 locations, but their cohorts, reference systems, and protocols differed from ours (Mariani et al., 2010;
386 Trojaniello et al., 2014; Laidig et al., 2021; Rampp et al., 2015). Direct performance rankings are therefore
387 inappropriate. Our study adds criterion validation in 32 older adults across single-task and graded dual-
388 task walking, with 10,472 paired strides and both RMSE and MAE reported. A multi-position study in
389 younger adults found greater single-stride variability at the heel than at other foot locations, especially
390 during fast walking (Küderle et al., 2022). A later heel-mounted validation also found greater spread
391 than a sensor embedded in the shoe, although mean errors remained small (Luo et al., 2024). Embedded
392 sensors may be preferable when footwear modification is practical. We chose a clip-on heel attachment
393 because it requires no shoe modification and can be applied quickly in clinical or home settings.

394 **Accuracy remained stable across cognitive load, while cognitive-status results warrant further**
395 **study.** Per-condition RMSE stayed between 3.7 and 4.0 cm even as cognitive load slowed gait by
396 16%. This mirrors a foot-mounted IMU validation in stroke and healthy adults that found no accuracy
397 decrement under irregular stepping (Ensink et al., 2023). We found no evidence of a condition effect
398 on participant-level MAE ($F(2.35, 68.27) = 0.67$, Greenhouse–Geisser corrected $p = .541$, $\eta_G^2 = 0.016$).
399 The 0.3 cm spread across condition RMSE values was modest, although the study was not designed as
400 an equivalence test. Stride-level RMSE was 3.65 cm in CN participants and 4.14 cm in CI participants.
401 Participant-level MAE was modestly higher in the CI group ($p = .046$). A sensitivity analysis restricted
402 to the twelve CI participants with MoCA scores of 18–25 returned 4.01 cm RMSE (Supplementary
403 Table S3). This subgroup analysis is exploratory and concerns MoCA screening categories rather than
404 clinical diagnoses. Diagnostic utility and longitudinal monitoring require prospective studies with formal
405 clinical classification and repeated measurements.

406 **The immediate application is supervised gait assessment when an instrumented walkway is im-**
407 **practical.** The heel clips require no footwear modification, and the pipeline adapts without participant
408 calibration. The stable error across task difficulty supports research and clinical evaluation during straight
409 walking passes. The present data do not support diagnostic use or unsupervised monitoring.

410 **The design comparisons supported the retroactive correction and internal mid-stance anchor.**
411 Disabling the retroactive gravity-residual correction increased bias from +0.3 to +2.2 cm and RMSE
412 from 3.9 to 4.9 cm (Supplementary Table S4). A small horizontal acceleration can remain after gravity
413 removal and accumulate when integrated across the stride. Correcting this residual reduced the offset
414 without a calibration walk or reference device. Heel-strike timing stayed inside the 150 ms validation
415 tolerance (Table 1). Replacing the internal mid-stance boundaries with GaitRite heel strikes did not
416 improve accuracy (Table 2). Bilateral validation recovered more matched strides than threshold-only
417 detection with similar accuracy. Its intended advantage during stops and turns will require validation
418 outside straight walkway passes (Mariani et al., 2010; Lin et al., 2025).

419 **The pipeline measured mean gait parameters more accurately than stride-to-stride variability.**
420 Gait speed, stride time, and cadence produced pooled ICC values of 0.90–0.96 (Table 4), consistent
421 with prior foot- and trunk-mounted IMU studies (Trojaniello et al., 2014; Washabaugh et al., 2017). In
422 contrast, pooled stride-length CV agreement was moderate (ICC 0.58), and the IMU overestimated CV
423 by 0.50 percentage points. Accurate mean stride length therefore did not guarantee equally accurate
424 stride-to-stride variability. Variability estimates should be interpreted cautiously until they are validated in
425 longer walking bouts with more strides per condition.

426 **Five design constraints limit generalisation of our results.** First, GaitRite ground truth covers only
427 supervised, straight strides across a 6.1 m walkway. The protocol did not test turns, uneven surfaces,
428 stops, footwear variation, long continuous bouts, or unsupervised activity. It therefore cannot establish
429 everyday-living accuracy or validate the bilateral detector's intended rejection of non-walking events.
430 Those questions require a longer-duration criterion such as synchronised video or wearable-compatible
431 motion capture. Second, participants walked independently with relatively typical gait patterns and the
432 sample skewed towards female (24 of 32 participants). Generalisation to severe gait impairments and sex-
433 specific accuracy claims remains untested. Third, cognitive classification relied on MoCA screening rather
434 than formal neuropsychological assessment. The subgroup findings concern screening categories, not
435 diagnostic discrimination or disease monitoring. Fourth, the retroactive correction adds one-stride latency
436 and prevents real-time estimation, although it does not affect offline applications. Fifth, single-session
437 data leave test-retest reliability and responsiveness to clinical change untested.

438 5 CONCLUSIONS

439 A calibration-free heel-mounted IMU pipeline estimated stride length in older adults to within 3.9 cm
440 RMSE (ICC 0.976) across single-task and graded dual-task walking. Per-condition RMSE ranged from
441 3.7 to 4.0 cm, and error was low in both MoCA-defined groups. Stride-length variability was less
442 accurate than mean stride length. These findings establish criterion validity for successfully synchronised,
443 supervised straight walkway passes. They do not establish diagnostic utility, longitudinal responsiveness,
444 or everyday-living accuracy. Independent validation should test other clinical cohorts, repeated sessions,
445 and unsupervised walking with turns, stops, varied footwear, and uneven surfaces.

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451 **COMPETING INTERESTS**

452 The authors declare that they have no competing interests.

453 **AUTHOR CONTRIBUTIONS**

454 **Pawel Kudzia** conceived the algorithm and analysis approach, designed the data-processing pipeline,
455 performed the data analysis, prepared figures and tables, drafted the manuscript, and approved the final
456 draft.

457 **Markus von Hacht** contributed to data collection, reviewed and edited drafts of the article, and approved
458 the final draft.

459 **Maryam Sheikhi** contributed to data collection and analysis, reviewed and edited drafts of the article,
460 and approved the final draft.

461 **Drew Commandeur** contributed to development of the data-collection and synchronisation methods,
462 reviewed and edited drafts of the article, and approved the final draft.

463 **Marc Klimstra** contributed to development of the data-collection and synchronisation methods, reviewed
464 and edited drafts of the article, and approved the final draft.

465 **Sandra Hundza** conceived and designed the experiments, supervised data collection, supervised the
466 analysis, reviewed and edited drafts of the article, and approved the final draft.

467 **HUMAN ETHICS**

468 The University of Victoria Human Research Ethics Board approved this study (protocol H22-00451). All
469 participants provided written informed consent.

470 DATA AVAILABILITY

471 We will deposit the de-identified stride-level data and the analysis code that support the findings of this
472 study in a public repository (Zenodo) and release them under an open licence upon publication. We will
473 additionally host the analysis code on GitHub, and we will provide a persistent DOI and the repository
474 URL at that time.

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